Felipe Fazio da Costa

Computer Engineering Student | Embedded Systems & Virtualization | Internship, Summer 2027

Metz, France | felipefazio.costa@gmail.com | GitHub | LinkedIn | Portfolio

Profile

Computer Engineering student in a double-degree program between Instituto Mauá de Tecnologia (Brazil) and CentraleSupélec (Metz, France). Hands-on experience building QEMU-based virtual platforms for aerospace instruments (ARM, LEON3) with the German Aerospace Center (DLR) and the Paris Observatory, and developing real-time telemetry for a Shell Eco-marathon vehicle. Comfortable across the hardware/software boundary, in C/C++ and Python, in international teams. Available for a 3-month internship in summer 2027 (internship agreement via CentraleSupélec).

Education

CentraleSupélec | *Engineering Degree – Computer Science (Double Degree)* 2026–2028

Instituto Mauá de Tecnologia (IMT) | *Bachelor's in Computer Engineering* 2023–2028

Research & Professional Experience

Embedded Electronic Systems Center (NSEE) – IMT 2024–2026

Undergraduate Researcher

- Designed and developed QEMU-based virtual platforms to emulate a spectrograph instrument for aerospace missions, including VERITAS and EnVision.
- Built emulation of three different processors with the Paris Observatory, enabling software testing without physical prototypes.
- Worked with DLR engineers on hardware emulation and validation in an international, multidisciplinary team.

Eco Mauá – Instituto Mauá de Tecnologia 2023–2026

Electrical Engineering Member – Student Team

- Contributed to the electrical system of a high-efficiency vehicle for the Shell Eco-marathon, where the team finished 8th.
- Developed real-time telemetry and monitoring with InfluxDB and Grafana, tracking 5–6 sensor channels.
- Integrated embedded electronics, instrumentation, and data acquisition with the vehicle's systems.

Selected Projects

VANESSA – CubeSat Instrument Emulator QEMU / ARM / Embedded

Virtual CubeSat instrument on three ARM-based virtual platforms (camera acquisition, image correction, telemetry) to validate the design before hardware.

QEMULA – LEON3 Virtual Instrument QEMU / LEON3 / Aerospace / Docker

QEMU-based LEON3 emulator to virtualize and test a spectrograph instrument for space-mission research.

SGPPF – Research Project Management Platform React / Node.js / Supabase / Vercel

Web platform for research projects, funding, and documentation, built by a team of 3 in 5 days.

HP Hackathon – Winning Team Python / Supply Chain / Hardware

Hackathon challenge on improving the supply chain of HP's factories in Brazil. With my team, developed a system to keep the production line running almost continuously.

NSEE Research Infrastructure Python / Data / Server / Docker

Developed a local environment to build and run CI/CD workflows.

Technical Skills

Languages	C, C++, Python, Java, JavaScript, SQL, MATLAB
Embedded	QEMU, ARM, LEON3, SPI, microcontrollers, telemetry, data acquisition
Web & Data	React, Node.js, Vite, Supabase, Vercel, Grafana, InfluxDB, survival analysis
Tools & Cloud	Linux, Git/GitHub, Docker, Podman, AWS, Google Cloud, EasyEDA

Languages

Portuguese (Native) | English (Advanced, C1 Cellep) | French (Intermediate, B1 DELF & TCF) | Spanish (Beginner)

Volunteer Experience

Instituto de Educação Beatíssima Virgem Maria 2017–2022

Supported NGOs and donation campaigns, including for people experiencing homelessness, and took part in activities in daycares and nursing homes.